3.2 Quickstart: AirportDB Analytics

The *MySQL HeatWave airportdb Quickstart* shows how to import data into the DB System using the MySQL Shell Dump Load utility, loading data into MySQL HeatWave Cluster using Auto Parallel Load, and running queries.

For an online workshop that demonstrates MySQL HeatWave using the `airportdb` sample database, see Turbocharge Business Insights with MySQL HeatWave Service and MySQL HeatWave.

This quickstart contains the following sections:

- AirportDB Requirements
- Download AirportDB Data
- Load AirportDB into DB system
- Load AirportDB into MySQL HeatWave Cluster
- Run AirportDB Queries
- Additional AirportDB Queries
- Unload AirportDB Tables

AirportDB Requirements

- Review the requirements to setup and use MySQL HeatWave.
- Confirm you have the latest version of MySQL Shell installed. To learn more about downloading and installing MySQL Shell, see Installing MySQL Shell.
- You will need to connect to the DB System in one of the following ways:
  - Through a VPN
  - Through a compute instance

  To learn more about connecting to your DB system, see Networking.

Download AirportDB Data

Download the `airportdb` sample database. The `airportdb` sample database is provided for download as a compressed **tar** or Zip archive. The download is approximately 640 MBs in size.

To run these commands, make sure you have the `wget` utility installed. Alternatively, you can download the files by pasting the URL in a browser window.

```
$> wget https://downloads.mysql.com/docs/airport-db.tar.gz
```

or

```
$> wget https://downloads.mysql.com/docs/airport-db.zip
```

Load AirportDB into DB system

Depending on how you connect to your DB system, follow the appropriate steps to load the `airportdb` database into the DB system. To learn more about connecting to your DB system, see Networking.

To load the data, you will need to use the MySQL Shell Dump Loading utility. See MySQL Shell Dump Loading Utility.

VPN Connection

If you connect to the DB system through a VPN connection, you can load `airportdb` locally from your computer.

To load the `airportdb` database through a VPN connection:

1. Unpack the compressed **tar** or Zip archive you downloaded previously. It creates a single directory named `airport-db`, which contains the data files.

   ```
   $> tar xvzf airport-db.tar.gz
   ```

   or

   ```
   $> unzip airport-db.zip
   ```

2. Copy the file path to the unpacked `airport-db` folder. For example, on Windows it might be `C:\\Temp\\airport-db`, or on Mac it might be `/Users/johnsmith/airport-db`.

3. From a terminal window, start MySQL Shell and connect to the DB System IP address. You can review the IP address (private IP address) and other connection details in the *Connections* tab for

the DB system. For additional information about connecting to a DB System, see Connecting to a DB System in the *MySQL HeatWave on OCI Service Guide*.

```
$> mysqlsh Username@DBSystem_IP_Address_or_Host_Name
```

4. Enable `js` mode.

```
mysql-sql> \js
```

5. Load the `airportdb` database into the DB system using MySQL Shell Dump Loading Utility. In the command, replace `airport-db` with the file path you copied in the previous step.

```
mysql-js> util.loadDump("file path to the unpacked airport-db folder", {thread
```

The command uses the following options:

- `threads`: Specifies the number of parallel threads to use to upload chunks of data to the target MySQL instance.
- `deferTableIndexes`: This option is set to "all", so it defers the creation of secondary indexes until after the table data is loaded. This can reduce loading time. If you intend to use `airportdb` with only MySQL HeatWave, which does not use secondary indexes, you can avoid creating secondary indexes by specifying the `loadIndexes: "FALSE"` option instead of `deferTableIndexes: "all"`. For more information about MySQL Dump Load options, see Dump Loading Utility.
- `ignoreVersion`: This option is enabled, which imports the dump even if the major version number of the MySQL instance from which the data was dumped is non-consecutive to the major version number of the MySQL instance to which the data will be uploaded.

Compute Instance Connection

If you connect to the DB system through a compute instance, you first need to upload the `airportdb` database to the compute instance before running the MySQL Shell Dump Loading Utility.

To load the `airportdb` database through a compute instance connection:

1. In the terminal window, go to the local directory containing the `airportdb` files.
2. Upload the contents of the `airport-db` folder to the appropriate directory of the compute instance.

```
$> scp -v -i ssh-key.key airport-db.zip opc@ComputeInstancePublicIP:/home/opc/
```

Replace the following:

- ***ssh-key.key***: The full file path to the SSH key file (.key) for the compute instance.
- ***airport-db.zip***: The full file path to the `airport-db.zip` file on your device.
- ***opc@ComputeInstancePublicIP***: The appropriate username and public IP for the compute instance.
- ***/home/opc/***: The appropriate file path to upload the airport-db folder to in the compute instance.

3. Once the upload successfully completes, SSH into the Compute instance using the public IP address of the Compute instance.

```
$> ssh -i ssh-key.key opc@computeInstancePublicIP
```

Replace the following:

- ***ssh-key.key***: The full file path to the SSH key file (.key) for the compute instance.
- ***opc@ComputeInstancePublicIP***: The appropriate username and public IP for the compute instance.

4. Unpack the compressed **tar** or Zip archive you downloaded previously. It creates a single directory named `airport-db`, which contains the data files.

```
$> tar xvzf airport-db.tar.gz
```

or

```
$> unzip airport-db.zip
```

5. From a terminal window, start MySQL Shell and connect to the DB System IP address. You can review the IP address (private IP address) and other connection details in the *Connections* tab for the DB system. For additional information about connecting to a DB System, see Connecting to a DB System in the *MySQL HeatWave on OCI Service Guide*.

```
$> mysqlsh Username@DBSystem_IP_Address_or_Host_Name
```

6. Enable `js` mode.

```
mysql-sql> \js
```

7. Load the `airportdb` database into the DB system using MySQL Shell Dump Loading Utility. If needed, replace `airport-db` to the appropriate file path in the compute instance.

```
mysql-js> util.loadDump("airport-db", {threads: 16, deferTableIndexes: "all",
```

The command uses the following options:

- `threads`: Specifies the number of parallel threads to use to upload chunks of data to the target MySQL instance.
- `deferTableIndexes`: This option is set to "all", so it defers the creation of secondary indexes until after the table data is loaded. This can reduce loading time. If you intend to use `airportdb` with only MySQL HeatWave, which does not use secondary indexes, you can avoid creating secondary indexes by specifying the `loadIndexes: "FALSE"` option instead of `deferTableIndexes: "all"`. For more information about MySQL Dump Load options, see Dump Loading Utility.
- `ignoreVersion`: This option is enabled, which imports the dump even if the major version number of the MySQL instance from which the data was dumped is non-consecutive to the major version number of the MySQL instance to which the data will be uploaded.

Load AirportDB into MySQL HeatWave Cluster

To load the `airportdb` from the DB System into MySQL HeatWave Cluster:

> **Note**
>
> For MySQL HeatWave on AWS, load data into MySQL HeatWave Cluster using the MySQL HeatWave Console. See Manage Data in MySQL HeatWave with Workspaces in the *MySQL HeatWave on AWS Service Guide*.

1. From a terminal window, start MySQL Shell and connect to the DB System IP address. You can review the IP address (private IP address) and other connection details in the *Connections* tab for

the DB system.

```
$> mysqlsh Username@DBSystem_IP_Address_or_Host_Name
```

2. Run the following Auto Parallel Load command to load the `airportdb` tables into MySQL HeatWave Cluster.

```
mysql-sql> CALL sys.heatwave_load(JSON_ARRAY('airportdb'), NULL);
```

For information about the Auto Parallel Load utility, see Section 4.2.5, “Load Data Using Auto Parallel Load”.

Run AirportDB Queries

After airportdb sample database tables are loaded into the MySQL HeatWave Cluster, queries that qualify are automatically offloaded to the MySQL HeatWave Cluster for accelerated processing. To run queries:

> **Note**
>
> For MySQL HeatWave on AWS, run queries from the *Query Editor* in the MySQL HeatWave Console. See Running Queries in the *MySQL HeatWave on AWS Service Guide*.

1. From a terminal window, start MySQL Shell and connect to the DB System IP address. You can review the IP address (private IP address) and other connection details in the *Connections* tab for the DB system.

```
$> mysqlsh Username@DBSystem_IP_Address_or_Host_Name
```

2. Change to the `airportdb` database.

```
mysql-sql> USE airportdb;
Default schema set to `airportdb`.Fetching table and column names from `airpor
auto-completion... Press ^C to stop.
```

3. Before running a query, use `EXPLAIN` to verify that the query can be offloaded to the MySQL HeatWave Cluster. For example:

```
mysql-sql> EXPLAIN SELECT booking.price, count(*)
                 FROM booking
                 WHERE booking.price > 500
                 GROUP BY booking.price
                 ORDER BY booking.price LIMIT
                 10\G
*************************** 1. row ***************************
           id: 1
  select_type: SIMPLE
        table: booking
   partitions: NULL
         type: ALL
possible_keys: NULL
          key: NULL
      key_len: NULL
          ref: NULL
         rows: 54081693
     filtered: 33.32999801635742
        Extra: Using where; Using temporary; Using filesort; Using secondary e
```

If the query can be offloaded, the `Extra` column in the `EXPLAIN` output reports `Using secondary engine RAPID`.

4. After verifying that the query can be offloaded, run the query and note the execution time.

```
mysql-sql> SELECT booking.price, count(*)
                 FROM booking
                 WHERE booking.price > 500
                 GROUP BY booking.price
                 ORDER BY booking.price
                 LIMIT 10;
+--------+----------+
| price  | count(*) |
+--------+----------+
| 500.01 |      860 |
| 500.02 |     1207 |
| 500.03 |     1135 |
| 500.04 |     1010 |
| 500.05 |     1016 |
| 500.06 |     1039 |
| 500.07 |     1002 |
| 500.08 |     1095 |
| 500.09 |     1117 |
| 500.10 |     1106 |
+--------+----------+
10 rows in set (0.0537 sec)
```

5. To compare the MySQL HeatWave Cluster execution time with DB System execution time, disable the `use_secondary_engine` variable to see how long it takes to run the same query on the DB System; for example:

```
mysql-sql> SET SESSION use_secondary_engine=OFF;
mysql-sql> SELECT booking.price, count(*)
                 FROM booking
                 WHERE booking.price > 500
                 GROUP BY booking.price
                 ORDER BY booking.price
                 LIMIT 10;
+--------+----------+
| price  | count(*) |
+--------+----------+
| 500.01 |      860 |
| 500.02 |     1207 |
| 500.03 |     1135 |
| 500.04 |     1010 |
| 500.05 |     1016 |
| 500.06 |     1039 |
| 500.07 |     1002 |
| 500.08 |     1095 |
| 500.09 |     1117 |
| 500.10 |     1106 |
+--------+----------+
10 rows in set (9.3859 sec)
```

For other `airportdb` sample database queries that you can run, see Additional AirportDB Queries. For more information about running queries, see Section 5.4, “Run Queries”.

Additional AirportDB Queries

This topic provides additional airportdb queries that you can run to test the MySQL HeatWave Cluster.

- Query 1: Number of Tickets > $500.00, Grouped By Price

```
mysql-sql> SELECT booking.price, count(*)
      FROM booking
      WHERE booking.price > 500
      GROUP BY booking.price
      ORDER BY booking.price
      LIMIT 10;
```

- Query 2: Average Age of Passengers By Country, Per Airline

```
mysql-sql> SELECT airline.airlinename, AVG(datediff(departure,birthdate)/365.2
      FROM booking, flight, airline, passengerdetails
      WHERE booking.flight_id=flight.flight_id
      AND airline.airline_id=flight.airline_id
      AND booking.passenger_id=passengerdetails.passenger_id
      AND country IN ("SWITZERLAND", "FRANCE", "ITALY")
      GROUP BY airline.airlinename
      ORDER BY airline.airlinename, avg_age
      LIMIT 10;
```

- Query 3: Most Tickets Sales by Airline for Departures from US Airports

```
mysql-sql> SELECT airline.airlinename, SUM(booking.price) as price_tickets, co
      FROM booking, flight, airline, airport_geo
      WHERE booking.flight_id=flight.flight_id
      AND airline.airline_id=flight.airline_id
      AND flight.from=airport_geo.airport_id
      AND airport_geo.country = "UNITED STATES"
      GROUP BY airline.airlinename
      ORDER BY nb_tickets desc, airline.airlinename
      LIMIT 10;
```

Unload AirportDB Tables

`airportdb` tables can be unloaded from MySQL HeatWave Cluster using the following statements:

> **Note**
>
> For MySQL HeatWave on AWS, unload data from MySQL HeatWave Cluster using the MySQL HeatWave Console. See Manage Data in MySQL HeatWave with Workspaces in the *MySQL HeatWave on AWS Service Guide*.

```
mysql-sql> USE airportdb;

ALTER TABLE booking SECONDARY_UNLOAD;
ALTER TABLE flight SECONDARY_UNLOAD;
ALTER TABLE flight_log SECONDARY_UNLOAD;
ALTER TABLE airport SECONDARY_UNLOAD;
ALTER TABLE airport_reachable SECONDARY_UNLOAD;
ALTER TABLE airport_geo SECONDARY_UNLOAD;
ALTER TABLE airline SECONDARY_UNLOAD;
ALTER TABLE flightschedule SECONDARY_UNLOAD;
ALTER TABLE airplane SECONDARY_UNLOAD;
ALTER TABLE airplane_type SECONDARY_UNLOAD;
```

```
ALTER TABLE employee SECONDARY_UNLOAD;
ALTER TABLE passenger SECONDARY_UNLOAD;
ALTER TABLE passengerdetails SECONDARY_UNLOAD;
ALTER TABLE weatherdata SECONDARY_UNLOAD;
```

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